

Vedansh Somani

Undergraduate Student

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Profile

Robotics & AI undergraduate specializing in machine learning, reinforcement learning, and autonomous systems. Experienced in developing deep learning models, RL agents, and ROS2-based robotic systems, with additional experience building full-stack applications.

Education

BE - Robotics and AI 09/2023 – Present
Dayananda Sagar College of Engineering Bengaluru, India

Professional Experience

STEM Trainer 04/2024 – 04/2025
Beylabee Bengaluru, India

- Delivered hands-on STEM and programming workshops, guiding students through robotics, electronics, and software-based projects.
- Designed and explained technical concepts through practical demonstrations, improving student understanding of engineering and computational principles.

Skills

Machine Learning and Reinforcement Learning • PyTorch • PPO • DQN (Deep Q-Learning) • Stable-Baselines3 • Neural Networks • Deep Learning • Reward Shaping • Curriculum Learning	Programming Languages • Python • C++ • HTML/CSS • JavaScript • Nodejs • React • Flutter
Robotics and Autonomous Systems • ROS2 • Nav2 • SLAM • Motion Planning • Robot Control • MuJoCo	Embedded Systems and Tools • ESP32 • Arduino • Microcontrollers • Git • Linux • TensorBoard

Projects

STRIDE v1 - RL Based Quadruped Locomotion (Unitree Go2) 06/2026 – Present
Python, MuJoCo, PPO, PyTorch

- Developed a PPO-based locomotion framework for a simulated Unitree Go2 quadruped using MuJoCo, custom environments, and PD joint control.

- Designed a PMTG-based gait architecture with foot trajectory generation, residual joint control, and per-leg phase modulation.
- Trained locomotion policies for stable trotting and disturbance recovery through reward shaping and quantitative policy evaluation.

Autonomous Mobile Robot

02/2026 – 05/2026

ROS2, Nav2, SLAM, LiDAR

- Developed an autonomous mobile robot using ROS2, LiDAR-based SLAM, and Nav2 for localization and navigation.
- Integrated SLAM, localization, path planning, and obstacle avoidance pipelines on a differential-drive platform.

Chain Reaction + RL

05/2026 – 05/2026

Python, PyTorch, Deep Q-Learning

- Recreated the Chain Reaction board game in Python/Pygame and developed a DQN agent with CNN-based state representation using PyTorch.
- Built an RL training pipeline with curriculum learning, reward shaping, and heuristic-guided action filtering, achieving 57% win rate against a greedy baseline.

4WD Rover

07/2025 – 08/2025

ESP32, Embedded Systems

- Built a wireless 4WD rover using ESP32 and Cytron motor control with multiple input modes including Bluetooth, web interface, and RF controller.
- Designed embedded control logic for real-time motor actuation and remote operation.

Chat Application

12/2025 – 02/2026

Flutter, Node.js, Socket Programming

Developed a full-stack mobile chat application with real-time socket communication, backend services, database integration, and encrypted authentication.

Jarvis-Lite Assistant

01/2026 – 02/2026

Built a voice-controlled assistant capable of command recognition and task execution through a rule-based permission system.

Courses

Neural Networks - Zero to Hero

06/2026 – Present

Andrej Karpathy

Bengaluru

Implemented neural networks from scratch, covering backpropagation, optimization, MLPs, CNNs, and transformer architectures using PyTorch.

Data Analysis with Python

04/2026 – 04/2026

IBM Developer Skills Network

Bengaluru

Covered data preprocessing, exploratory data analysis, statistical analysis, and visualization using Python libraries including NumPy, Pandas, and Matplotlib.

Python 101 for Data Science

04/2026 – 04/2026

IBM Developer Skills Network

Bengaluru

Covered Python-based data workflows including data manipulation, analysis, and scripting for data science applications.